

WASSWA WILSON

Full-Stack Software Developer

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English (Fluent), Luganda (Native)

PROFESSIONAL SUMMARY

I am a full-stack developer and data analyst with hospital, programme management, and research experience. I am the Founder and Developer of HOSSPI (<https://hosspi.com/>), a hospital management platform connecting patient records, clinical workflows, billing, biomedical engineering, and live analytics. My software work also includes Shulekeeper, endoscopy reporting, Wekebere, and an AI reception agent deployed at a Texas law firm. As Biomedical Programs Manager at Gould Family Foundation (GFF), I led biomedical engineers and technicians across Uganda, Kenya, Tanzania, Burundi, Malawi, Somaliland, and Rwanda. At IHK, I developed oxygen-consumption models and maintained compliance evidence; at UVRI, I analysed laboratory data and built Python research pipelines. My current FCHIP role at FairBanks combines software development with Python and SQL analysis, data-quality checks, and reporting dashboards.

TECHNICAL SKILLS

Web and mobile: Next.js, React, TypeScript, Tailwind CSS, React Native, Expo, HTML/CSS, Jest, Playwright

Backend and cloud: Node.js, Express, Convex, Better Auth, MySQL, Prisma, PHP (CodeIgniter, Laravel), Java, AWS, Azure, Vercel, EAS, Bunny.net

Integrations and delivery: REST APIs, JWT, RBAC, Resend, WhatsApp messaging, CI/CD, Git

Other: AI agents and ML tools; hospital and clinic software; Cursor, OpenAI, Claude

SOFTWARE EXPERIENCE

Lead Software Developer and Data Analyst, FairBanks Medical Centre (FCHIP)

Current

- Lead FCHIP development and maintain FairBanks hospital and clinic software supporting community health and clinical workflows.
- Analyse FCHIP health data in Python and SQL, define indicators and data-quality checks, and build dashboards for management reporting.
- Work across web and mobile with Next.js, TypeScript, Tailwind, React Native/Expo, Convex, auth, messaging, CDN, and cloud deploy, including storage and data sync.

SOFTWARE PROJECTS

HOSSPI (hospital management platform)

Founder and Developer | Current | <https://hosspi.com/>

- Founded and developed a platform connecting shared patient records, clinical, laboratory and pharmacy workflows, billing and finance, and biomedical engineering.
- Integrated role-based permissions, multi-facility management, and live analytics.

AI reception agent (law firm, Texas, USA)

Built and deployed; in active use

- Built and deployed an AI reception agent in active use at a Texas law firm, handling incoming calls, appointment booking and rescheduling, and client follow-up.

Shulekeeper (school information system)

Assistant Software Developer, 2021-2024

- Helped build the web app with React/Next.js and TypeScript, Laravel on the backend, MySQL, and AWS hosting.
- Gathered requirements, built UI components and integrations, tested and upgraded the system, trained users, and wrote manuals.

Endoscopy reporting software

Lead Software Developer, St. Catherine Hospital and St. Francis Hospital, 2017-2019

- Led the build of a medical image capture and reporting tool for endoscopy (JavaFX UI, MySQL database).
- Moved old data into the new system, installed it on hospital machines, and fixed issues as they came up.

Wekebere (fetal heart-rate monitoring)

Electronics Developer and Programmer, 2017-2024

- Built an Android app in Java that talks to Arduino hardware for fetal heart-rate checks in late pregnancy, with Azure in the setup.
- Worked out how the phone app and the hardware should connect and share data.

HOSPITAL AND BIOMEDICAL WORK

Biomedical Programs Manager, Gould Family Foundation (GFF)

Aug 2024 - Feb 2025

- Led biomedical engineers and technicians across Uganda, Kenya, Tanzania, Burundi, Malawi, Somaliland, and Rwanda; coordinated equipment inventory, planning, assessment, technical advisory, procurement, installation, and commissioning.
- Trained biomedical engineers and technicians and clinical users, including doctors, nurses, and apprentices, on equipment use and care; upheld healthcare technology standards.
- Led the NICU upgrade at Mama Lucy Kibaki Hospital, Nairobi, Kenya, and developed standard operating procedures (SOPs) for supported equipment in partner facilities.

Biomedical Manager, International Hospital Kampala (IHK)

Jan 2020 - Jan 2024

- Led IHK upgrades of theatre, laboratory, NICU, and oxygen plant, and installation of a 64-slice CT scanner and digital X-ray; supported ICU equipment projects.
- Developed oxygen-consumption models for capacity and stock planning; maintained preventive maintenance schedules, service records, and COHSASA compliance evidence.

Biomedical Engineer, Norvik Hospital Ltd

Apr 2019 - Jan 2020

- Serviced diagnostic and monitoring equipment and helped with installs, calibration, and preventive maintenance.

EARLIER ROLES

Research Intern, Uganda Virus Research Institute

Dec 2018 - Apr 2019

- Analysed laboratory data, developed mathematical models and research pipelines in Python, and documented research-data quality.

Teaching Assistant, Makerere University, College of Health Sciences

Mar 2016 - Aug 2017

- Helped with biomedical engineering classes and lab sessions.

EDUCATION

University of Washington

Certificate in Leadership and Management in Health, 2022 (A+)

Makerere University

BSc Biomedical Engineering (Second Upper Honours), 2012-2017. Key courses: ICT, OOP, Software Engineering, Database Systems

Green Bridge School of Open Technology

Advanced Java Programming, 2016 (A+)

REFEREES

Racheal Nabukeera - Director and Founder, FairBanks Medical Centre - nracheal017@gmail.com / +256 772 849 258

Eng. Richard Ssejongo - Biomedical Engineer, St. Francis Hospital Nsambya - +256 753 818 754 / +256 777 132 489

Dr. Annet Khingi - Senior Radiologist and Administrator, Mengo Hospital - +256 772 592 771 / +256 701 592 771